NumPy: The Absolute Basics for Beginners

1 . What is NumPy?
2 . Open source Python library for numerical computing.
3 . Provides N-dimensional arrays (ndarray) and fast operations.

4 . Best for large homogeneous data.

5 . Importing NumPy

```
import numpy as np
```

6 . Arrays

7 . Arrays store data in grid-like structures of same data type.
8 . Fixed size and rectangular shape.

```
a = np.array([1, 2, 3, 4, 5, 6])
a.ndim   # Number of dimensions
a.shape  # Shape of array
a.size   # Total number of elements
a.dtype  # Data type
```

- Indexing & slicing:

```
a[0] = 10
b = a[3:]
b[0] = 40  # modifies original 'a'
```

- 2 D arrays:

```
a = np.array([[1,2,3,4], [5,6,7,8], [9,10,11,12]])
a[1,3]  # element at row 1, column 3
```

- Creating basic arrays

```
np.zeros(2)
np.ones(2)
np.empty(2)
np.arange(2, 9, 2)
```

```
np.linspace(0, 10, num=5)
np.ones(2, dtype=np.int64)
```

- Adding, removing, and sorting elements

- Sorting: np.sort(arr)
- Concatenate: np.concatenate((a,b))

- Remove: Use indexing to select what to keep

- Array attributes

- ndim : number of dimensions
- size : total elements
- shape: elements per axis

- dtype : element type

- Reshaping and adding axes

```
a.reshape(3,2)
a[np.newaxis, :]  # row vector
a[:, np.newaxis]  # column vector
np.expand_dims(a, axis=0 or 1)
```

- Indexing & slicing

```
a[a < 5]
a[(a > 2) & (a < 11)]
np.nonzero(a < 5)
```

- Stacking and splitting

```
np.vstack((a1,a2))
np.hstack((a1,a2))
np.hsplit(x, 3)
```

- Copies vs Views

- view(): shallow copy, modifies original

- copy(): deep copy

- Basic array operations

```
data + ones
data - ones
data * data
data / data
a.sum(axis=0 or 1)
```

- Broadcasting

```
data * 1.6
```

- Aggregation

```
data.max(), data.min(), data.sum(), data.mean(), data.std(), data.prod()
```

- Matrices

- Create: np.array([[1 , 2],[3 , 4]])
- Access: data[0 , 1], data[1 : 3]

- Aggregation: data.max(axis= 0 or 1)

- Transposing & reshaping

```
data.T
data.reshape(2,3)
arr.transpose()
```

- Reversing arrays

```
np.flip(arr)
np.flip(arr, axis=0 or 1)
```

- Flattening arrays

```
x.flatten()  # copy
x.ravel()    # view
```

- Random numbers

```
rng = np.random.default_rng()
rng.random(3)
rng.integers(low, high, size=(2,4))
```

- Unique items and counts

```
np.unique(a)
np.unique(a, return_index=True)
np.unique(a, return_counts=True)
np.unique(a, axis=0)
```

- Docstrings

```
help(np.array)
```